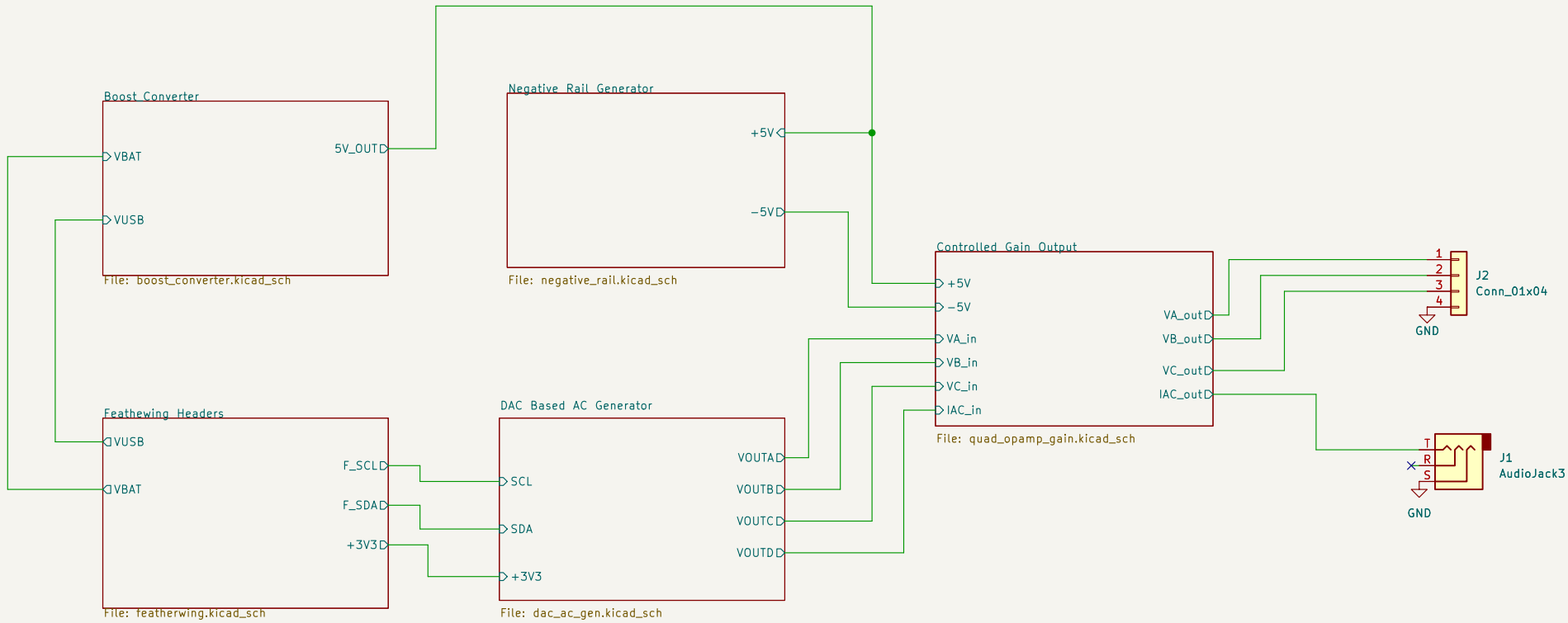
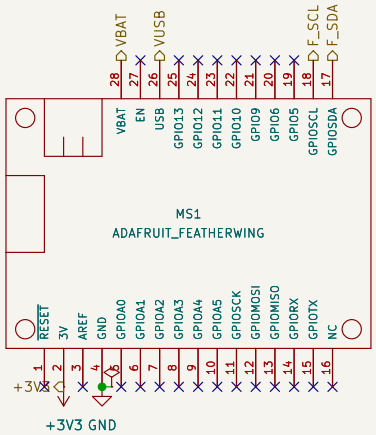



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**Title: 3-Phase Synth FeatherWing**

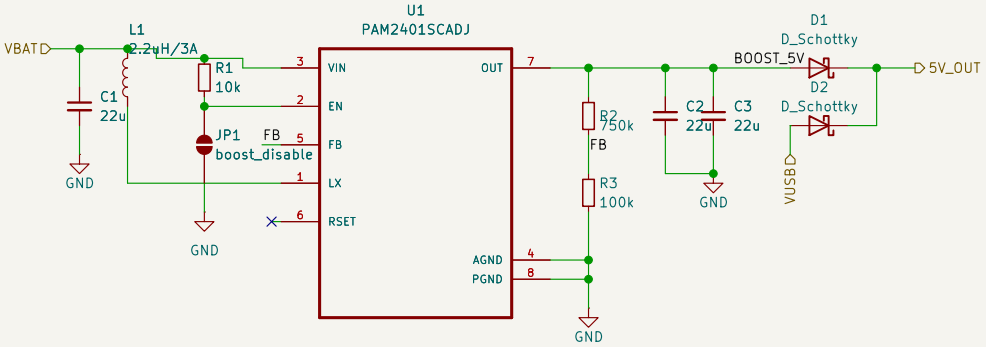
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| KiCad E.D.A. 10.0.3 | | Id: 1/6 |

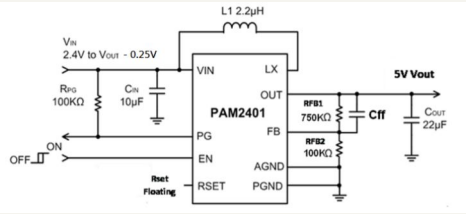



Sheet: /Feathewing Headers/
File: featherwing.kicad_sch

**Title: 3-Phase Synth FeatherWing**

| Size: A4 | Date: 2026-10-28 | **Rev: 0.4** |
|---|---|---|
| KiCad E.D.A. 10.0.3 | | Id: 2/6 |

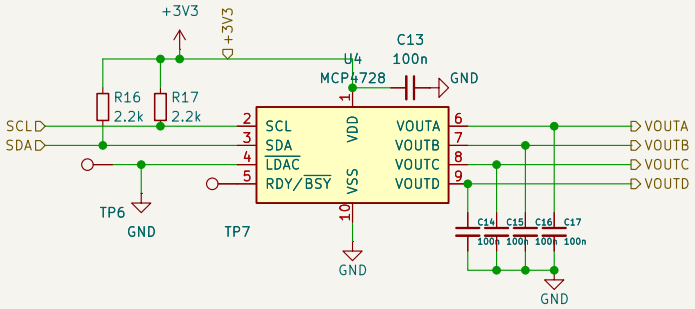


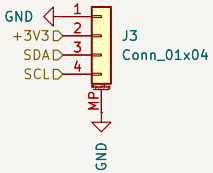



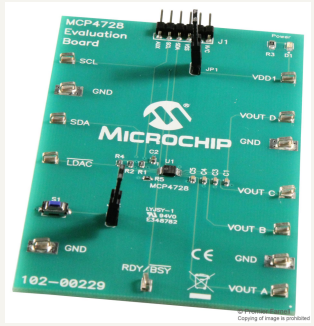

Sheet: /Boost_Converter/
File: boost_converter.kicad_sch

**Title: 3-Phase Synth FeatherWing**

| Size: A4 | Date: 2026-10-28 | **Rev: 0.4** |
|---|---|---|
| KiCad E.D.A. 10.0.3 | | Id: 3/6 |



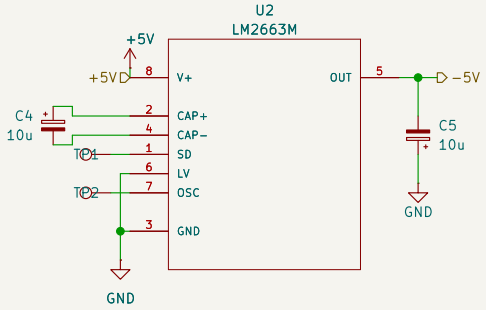



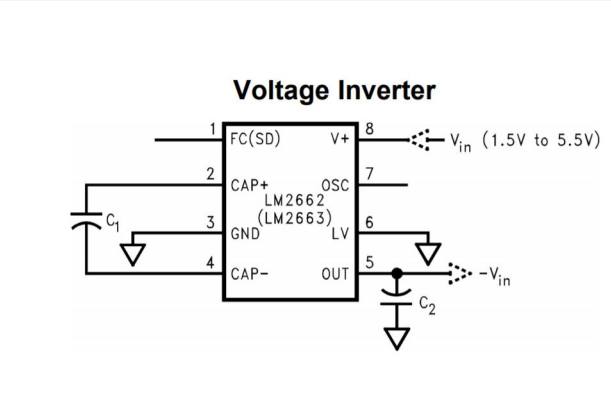




Sheet: /DAC Based AC Generator/
File: dac_ac_gen.kicad_sch

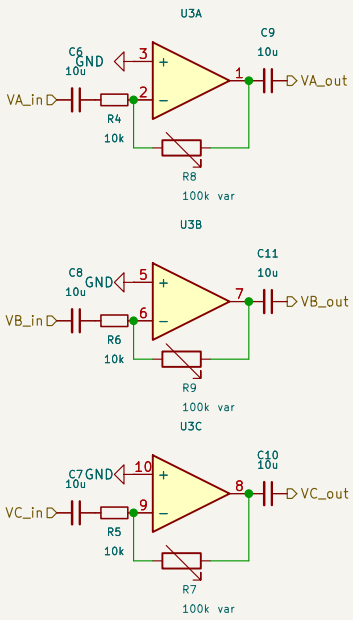
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|---|---|---|
| KiCad E.D.A. 10.0.3 | | Id: 4/6 |

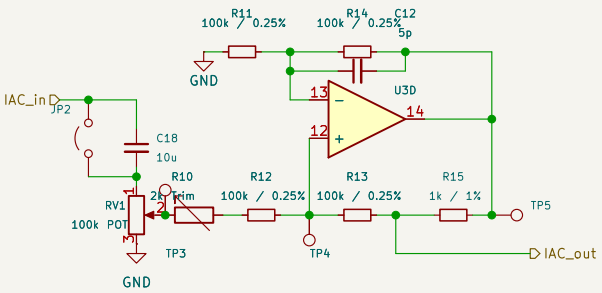



| | | |
|---|---|---|
| Sheet: /Negative Rail Generator/ | | |
| File: negative_rail.kicad_sch | | |
| **Title: 3-Phase Synth FeatherWing** | | |
| Size: A4 | Date: 2026-10-28 | **Rev: 0.4** |
| KiCad E.D.A. 10.0.3 | | Id: 5/6 |

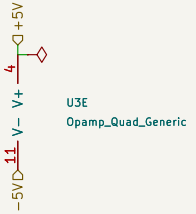



Pot controlled inverting amplifier




IAC_in is actually a voltage input used to control
the Howland Current Pump (improved with trim-pots).
The voltage can be tapped to control the current output.

Voltage controlled constant AC current-source




Dual-voltage opamp power rails

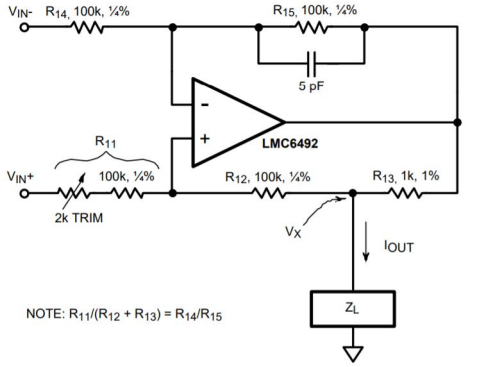



**Figure 6. The "Improved Howland" with Trim for $Z_{OUT}$**

| | | |
|---|---|---|
| Sheet: /Controlled Gain Output/ | | |
| File: quad_opamp_gain.kicad_sch | | |
| ***Title: 3-Phase Synth FeatherWing*** | | |
| Size: A4 | Date: 2026-10-28 | **Rev: 0.4** |
| KiCad E.D.A. 10.0.3 | | Id: 6/6 |